DiaTool-DPO: Multi-Turn DPO for Tool-Augmented LLMs

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

- Models TA-LLM interactions as a 5-state MDP with 3 query types
- Automatically constructs preference pairs without human labeling
- Achieves near-GPT-4o conversational performance on smaller open-source models

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Core Problem & Failure Modes

Tool-Augmented LLMs must control dialogue flow across three decisions:

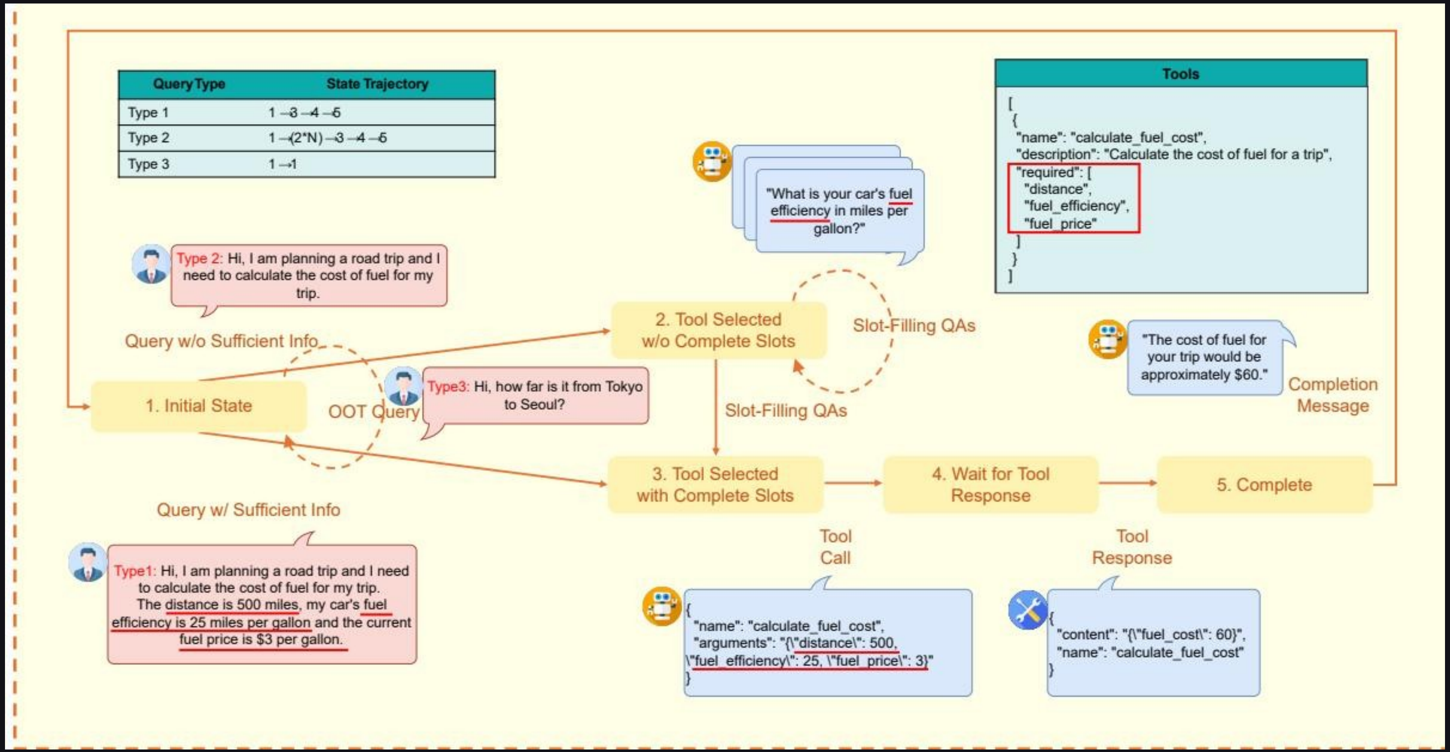
1. Ask follow-up questions when arguments are missing
2. Make a tool call when all required information is present
3. Reject tool calls when no suitable tool exists

SFT-only training on expert trajectories leads to:

- Hallucinated tool invocations — calling APIs with incomplete or fabricated slots
- Missing slot-filling QA — failing to ask clarifying questions for missing parameters
- Inability to reject out-of-scope requests — producing incorrect tool calls instead of denials

Why SFT is insufficient: it learns to imitate average behavior but lacks a mechanism to distinguish correct vs. incorrect dialogue flows.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process



1. Initial State

No history. Out-of-scope requests return a rejection message and remain here.

2. Tool Selected w/o Complete Slots

Tool identified but required parameters missing. Slot-filling QA occurs here.

3. Tool Selected w/ Complete Slots

Both tool selection and all argument values are determined.

4. Wait for Tool Response

Tool call executed, awaiting results.

5. Complete

Tool results received; completion message generated for the user.

Three Query Types Define Distinct State Trajectories

Query Type	State Trajectory	Example
Type 1	1 → 3 → 4 → 5	User provides all required arguments; tool is selected and called immediately.
Type 2	1 → (2×N) → 3 → 4 → 5	User misses some arguments; slot-filling QA repeats N times until complete.
Type 3	1 → 1	Requested functionality is unavailable; tool call must be rejected outright.

Automatic Paired Trajectory Construction Without Human Labels

Query	Chosen Trajectory	Rejected Trajectory	Learning Lesson
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination
Type 3	1→1	1→3→4	Tool call reject

Dataset spans ~11K pairs across Easy and Hard difficulty levels, generated entirely without human annotation.

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DiaTool-DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	**SFT + DiaTool-DPO**	**0.886**	**0.929**	**0.833**	**0.826**	**0.884**	**0.868**
Prop.-3.1B	DiaTool-DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	**SFT + DiaTool-DPO**	**0.743**	**0.871**	**0.833**	**0.826**	**0.765**	**0.818**
LLaMA-3-8B	DiaTool-DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	**SFT + DiaTool-DPO**	**0.857**	**0.929**	**0.917**	**0.913**	**0.905**	**0.904**

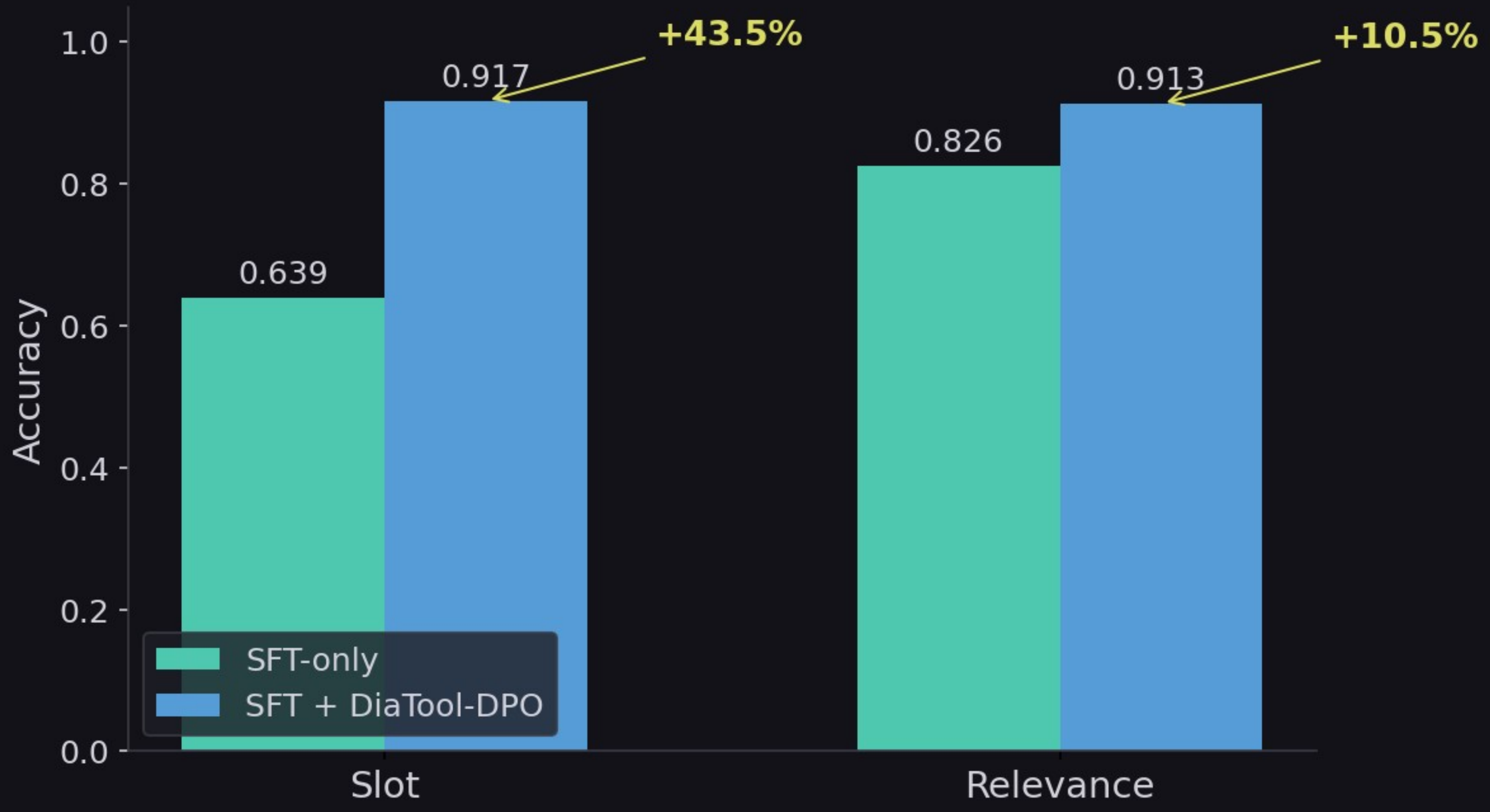
Bold teal rows highlight the best Macro Average per model. DPO alone underperforms; combined with SFT it yields the strongest balanced results.

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

SFT-only:
Slot = 0.639
Relevance = 0.826

SFT + DPO:
Slot = 0.917
Relevance = 0.913

Near GPT-4o performance on dialogue control



Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

1. MDP formulation enables systematic preference pair construction

Defining 5 internal states and 3 query types allows automatic generation of chosen/rejected trajectories without human annotation.

2. DPO complements SFT — it does not replace it

DiaTool-DPO-only performs poorly across all models; the gains appear only when DPO is applied on top of a strong SFT baseline.

3. Largest gains are in slot-filling and tool-call rejection

LLaMA-3-8B improves Slot accuracy by +43.5% and Relevance by +10.5%, reaching near-GPT-4o conversational performance.

4. Consistent improvement in Macro Average across model sizes

SFT + DiaTool-DPO achieves the best balanced performance on Prop.-8B, Prop.-3.1B, and LLaMA-3-8B.